



JUAN ROBERTO GARCIA GOMEZ

Data Engineer | MES & Manufacturing Analytics

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Summary

Economist and Industrial Engineer leading end-to-end **Data & Manufacturing Systems**. At Chunil México, We **built** a production-grade **web MES** (React, Node.js, PostgreSQL) that replaced fragmented Excel and is used daily across **Injection, Assembly, and EPC** (final inspection). I designed the relational data model and REST APIs; implemented in-portal KPIs and dashboards with **D3.js** (plan attainment, availability, performance from cycle times, quality, OEE); and **deliver Power BI reports** for production tracking and executive visibility. I **scaled the IT function from 1→3**, introduced **QR-based spare-parts inventory scanning** for Tooling/Assembly, and am leading the **IIoT/real-time** roadmap for injection machines and assembly stations **in partnership with an external consultant**. Previously at **General Motors (Finance/Supply Chain)**, I recovered **MX\$15.2M** in logistics claims and reduced case analysis time by **40%** through BI automation.

Skills

Data Engineering: PostgreSQL/SQL • Python (pandas) • Node.js/Express • REST APIs • Data Modeling • Docker • Git

Frontend & Viz: React • D3.js • Power BI (DAX, Power Query)

Manufacturing Systems: MES/ERP • IIoT (roadmap) • QR/traceability • Real-time data ingestion

Manufacturing Analytics & KPIs:

OEE (Availability, Performance, Quality) • Plan Attainment • Target vs Actual • Cycle-time performance • Availability/Downtime • Scrap & FPY • SPC

Domains: Injection molding • Logistic claims • Assembly • Inventory/warehouse

Experience



Chunil México

San Luis Potosí, MX

Data Engineer (MES & Manufacturing Analytics)

10/2024 - Present

Greenfield transformation at *Chunil México*, a **Korean automotive manufacturing company**, moving from manual spreadsheets to a production-grade web MES used daily across Injection, Assembly, and EPC.

- Replaced fragmented Excel with a unified platform (**React, Node.js, PostgreSQL**) and a single source of truth.
- Built the **relational data model** and **REST APIs**; implemented ETL and governance for master data (**parts, machines, molds, cycle times, targets, shifts**).
- Delivered **in-portal KPIs & dashboards with D3.js** (Plan Attainment — **Target vs Actual**, Availability/Downtime, Cycle-time Performance, Quality, OEE) and **Power BI** reports for production tracking and executive visibility.
- Built and maintain the **Data Consultant module** (self-service explorer) with multi-criteria filters and **one-click Excel (XLSX) exports** for audits and management.
- **Scaled IT from 1→3**; recruited and mentored a multidisciplinary team; introduced Git-based versioning, environments, and release practices.
- Deployed **QR-based spare-parts inventory scanning** for Tooling/Assembly, extending traceability within the same platform.
- Leading the **IIoT/real-time** roadmap with an external consultant (tag mapping, event model, ingestion) for injection machines and assembly stations.



Ramírez Valle & Asociados (on-site at General Motors SLP)

San Luis Potosí, MX

Claims, Loss & Recovery Coordinator (Supply Chain)

10/2023 - 2024

Coordinated logistics claims, loss recovery, and supplier negotiations for **General Motors San Luis Potosí (GM SLP)** plant.

- **Recovered MX\$15.2M** in logistics claims (~70% full/partial reimbursement) through data-driven root-cause analysis and negotiations.
- Managed **100+ investigations**; built **Power BI** dashboards that reduced case analysis time by **40%** and improved executive visibility.
- Led monthly cross-functional reviews; standardized documentation/action plans, improving time-to-resolution by **~30%**.
- Built internal **tracking tools** that streamlined claim intake, evidence, and follow-ups, increasing operational efficiency.

Education

	KeepCoding (Spain) Full Stack Web Development Bootcamp	San Luis Potosí, MX 11/2025 - Present
	Tecnológico de Monterrey Business Analytics	San Luis Potosí, MX 01/2025 - 04/2025
	KeepCoding (Spain) Bootcamp: Big Data, Artificial Intelligence & Machine Learning	San Luis Potosí, MX 06/2023 - 06/2024
	Instituto Mexicano Educativo de Seguros y Fianzas Diploma in Property and Casualty Claims	San Luis Potosí, MX 02/2024 - 11/2024
	Instituto de Investigación en Comunicación Óptica (IICO) Python for Data Science	San Luis Potosí, MX 05/2022 - 08/2022
	Universidad Autónoma de San Luis Potosí Bachelor's Degree in Economics	San Luis Potosí, MX 08/2018 - 07/2023
	Universidad Interamericana para el Desarrollo Bachelor's Degree in Industrial & Logistics Engineering	San Luis Potosí, MX 08/2014 - 07/2017

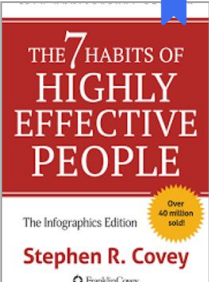
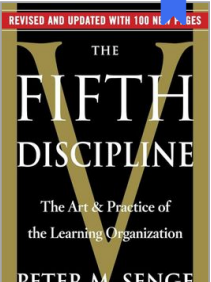
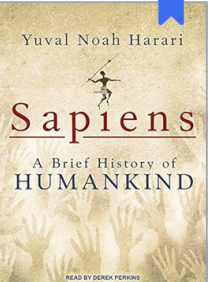
Selected Portfolio Projects

 Chunil MES Portal – Production, Quality & Analytics Production-grade MES with D3 KPIs (Plan, Availability, Performance, Quality, OEE), Data Consultant (filters + XLSX), QR spare-parts, and Power BI reporting; active IIoT roadmap. <i>Tech:</i> React · Node.js · PostgreSQL · REST APIs · D3.js · Power BI · XLSX (xlsx-js-style) View project on GitHub	 Tableau – Airbnb Host Dashboard Interactive map + slicers (zone, price, room type) and drill-downs (avg price by room type, availability, amenities, ratings). Packaged as .twbx for Tableau Desktop/Reader. <i>Tech:</i> Tableau. View project on GitHub	 SQL Analytics Portfolio Query library and data-modeling exercises showing joins, window functions, CTEs and ERDs; focus on operational/production datasets. <i>Tech:</i> PostgreSQL, BigQuery/ GCP. View project on GitHub
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Languages

English Proficient ●●●●● Spanish Native ●●●●●

Books

	The 7 Habits of Highly Effective People Stephen R. Covey
	The Fifth Discipline: The Art and Practice of the Learning Organization Peter M. Senge
	Sapiens: A Brief History of Humankind Yuval Noah Harari

Interests

 Camping & hiking  Video games  Electric guitar